

Questions

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Q1 - 2024 (05 Apr Shift 2)

A sonometer wire of resonating length 90 cm has a fundamental frequency of 400 Hz when kept under some tension. The resonating length of the wire with fundamental frequency of 600 Hz under same tension _____ cm

Q2 - 2024 (06 Apr Shift 2)

Two open organ pipes of lengths 60 cm and 90 cm resonate at 6th and 5th harmonics respectively. The difference of frequencies for the given modes is _____ Hz. (Velocity of sound in air = 333 m/s)

Q3 - 2024 (08 Apr Shift 1)

A closed and an open organ pipe have same lengths. If the ratio of frequencies of their seventh overtones is $\left(\frac{a-1}{a}\right)$ then the value of a is _____

Q4 - 2024 (08 Apr Shift 2)

A plane progressive wave is given by $y = 2 \cos 2\pi(330t - x)$ m. The frequency of the wave is :

- (1) 330 Hz
- (2) 660 Hz
- (3) 340 Hz
- (4) 165 Hz

Questions

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Answer Key

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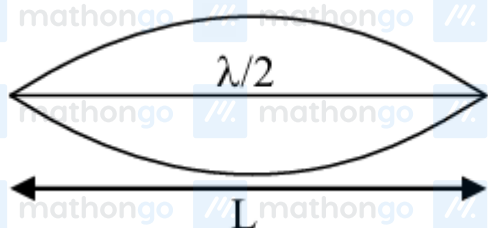
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Solutions

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Q1



$$f_0 = 400 \text{ Hz}; v = \sqrt{\frac{T}{\mu}} = \text{constant}$$

$$\frac{\lambda}{2} = L; \quad v = f_0 \lambda$$

$$\frac{\lambda}{2f_0} = L \Rightarrow v = 2Lf_0$$

$$L' = \frac{v}{2f'} = \frac{2Lf_0}{2f'} \\ = \frac{Lf_0}{f'} = \frac{90 \times 400}{600} = 60$$

Q2

The difference in frequency in open organ pipe =

$$f = \frac{nv}{2L}$$

$$\Delta f = \frac{6v}{2 \times 0.6} - \frac{5v}{2 \times 0.9}$$

$$v = 333 \text{ m/s}$$

$$\Delta f = 740 \text{ Hz}$$

Q3

For closed organ pipe

$$f_c = (2n+1) \frac{v}{4\ell} = \frac{15v}{4\ell}$$

For open organ pipe

$$f_o = (n+1) \frac{v}{2\ell} = \frac{8v}{2\ell}$$

$$\frac{f_c}{f_o} = \frac{15}{16} = \frac{a-1}{a}$$

$$\Rightarrow a = 16$$

Q4

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Solutions

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$$y = 2 \cos 2\pi(330t - x)m$$

$$y = A \cos(\omega t - kx)$$

$$\text{by comparing } \omega = 2\pi \times 330$$

$$2\pi f = 2\pi \times 330$$

$$f = 330$$

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